

1. Encircle the correct answer:

i) A digital signal has:
(a) Infinite values (b) Only two values
(c) Three values (d) Ten values

ii) Which gate gives output 1 only when all inputs are 1?
(a) OR gate (b) AND gate
(c) NOT gate (d) NOR gate

iii) XOR gate gives output 1 when inputs are:
(a) Same (b) Different
(c) Both 0 (d) Both 1

iv) Which phase comes first in SDLC?
(a) Testing (b) Requirement Analysis
(c) Coding (d) Design

v) In Agile, a short development cycle is called:
(a) Phase (b) Sprint
(c) Module (d) Release

vi) Which testing type does NOT require knowledge of code?
(a) White box testing (b) Black box testing
(c) Unit testing (d) Code review

vii) What device operates at the Network layer (Layer 3) of the OSI model?
(a) Hub (b) Switch
(c) Router (d) Repeater

viii) How many layers does the TCP/IP model have?
(a) 3 (b) 4
(c) 5 (d) 7

ix) UDP is preferred over TCP for video streaming because:
(a) UDP is more reliable (b) UDP is slower but more accurate
(c) UDP is faster and has lower delay (d) UDP guarantees delivery

x) Which OSI layer is responsible for routing packets between networks?
(a) Physical (b) Data Link
(c) Network (d) Transport

2. Fill in the Blanks:

i) An _____ gate gives output 1 only when both the inputs are same.
ii) The _____ gate is also called an inverter.
iii) In Agile, a short development cycle is called a _____.

SINDH TEXTBOOK BOARD

- Searching is the process of locating a specific element within a collection of data.
- Common searching algorithms include:
  - Linear Search
  - Binary Search
- Correctness is the property of an algorithm that ensures it produces the expected output for every valid input.
- Efficiency of an algorithm is measured by the amount of time it takes to execute (time complexity) and the amount of memory it uses (space complexity).
- Clarity refers to how easy an algorithm is to read, understand, and modify by others.



1. Encircle the correct answer

i) Which strategy involves ignoring the brand of a kettle when learning how to make tea?
(a) Decomposition (b) Pattern Recognition
(c) Abstraction (d) Sorting

ii) Bubble Sort is also commonly known as:
(a) Linear Sort (b) Ripple Sort
(c) Selection Sort (d) Binary Sort

iii) Which searching algorithm requires the data to be sorted before it can be used?
(a) Linear Search (b) Sequential Search
(c) Binary Search (d) Bubble Search

iv) What is the main disadvantage of Selection Sort?
(a) It uses extra RAM (b) It is difficult to code
(c) It must finish every loop even if the list is sorted
(d) It only works on alphabetical data

v) An algorithm is considered "correct" if:
(a) It runs very fast
(b) Its output matches expected outcomes for given inputs
(c) It uses minimal memory
(d) It is written in a specific programming language

vi) What does the abbreviation CT stand for?
(a) Computer Technology (b) Computational Thinking
(c) Central Terminal (d) Coding Technique

vii) Which operation is represented by MOD?
(a) Multiplication (b) Division
(c) Subtraction (d) Finding the remainder after division

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viii) In algorithms, what is the purpose of a PTR?
(a) To store text data (b) To perform calculations
(c) To track positions in an array (d) To display output

ix) What does INT represent in an algorithm?
(a) Internet (b) Integer
(c) Interval (d) Input

x) In the Binary Search algorithm, the abbreviations BEG, END, and MID represent:
(a) Input, Process, and Output (b) Beginning, End, and Middle positions
(c) First, Last, and Next values (d) Start, Stop, and Result

2. Fill in the Blanks

i) _____ is a step-by-step logical procedure to solve a problem.
ii) Sorting may be Ascending or _____.
iii) Algorithm is _____ and language-independent.
iv) _____ algorithm is designed to locate a specific element within list of data.
v) We measure any algorithm efficiency through _____ and _____ as resources.

3. Provide short answers to the following questions.

i) Define pattern recognition.
ii) Explain the concept of decomposition.
iii) What is meant by correctness in problem-solving?
iv) Why is efficiency important in problem-solving?
v) What do you understand by clarity in problem-solving?

4. Provide detailed answers to the following questions.

i) What is the difference between an algorithm and pseudocode?
ii) What is abstraction in an algorithm, and why is it used?
iii) Write the steps of the Selection Sort algorithm.
iv) Write the steps of the Binary Search algorithm.
v) Which searching algorithm is best for large datasets? Explain your answer.

Activities

i) Arrange the assembly queue according to students' heights.
ii) Arrange the names in the attendance register in descending alphabetical order.
iii) Find the name of your class monitor in the attendance register using the binary search technique.

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1. Encircle the correct answer:

i) Which is the correct statement to insert Comments in Python code?
(a) // This is comment (b) /* This is comment*/
(c) # This is comment (d) - - This is comment

ii) Which one is not a legal variable name?
(a) _myvar (b) my_var
(c) Myvar (d) my-var

iii) Which one is the correct way to write if statement in Python?
(a) if (condition): (b) if(condition);
(c) if (1): (d) none of these

iv) ___ is the operator can be used to compare two values?
(a) < > (b) = =
(c) >< (d) =

v) ___ loop is used when the number of iterations is known in advance
(a) While (b) Do while
(c) For (d) For each

vi) Which programming language is closest to binary (0s and 1s)?
(a) High Level language (b) Mid-Level Language
(c) Low Level language (d) Script Language

vii) Standard indentation in Python?
(a) 2 spaces (b) 4 spaces
(c) 6 spaces (d) 8 spaces

viii) __ data type does the input() function always return.
(a) Int (b) Float
(c) String (d) Bool

ix) ___ statement exits a loop immediately
(a) Pass (b) Continue
(c) Break (d) Exit

x) ___ bitwise operator returns 1 only when BOTH bits are 1?
(a) AND (b) OR
(c) NOT (d) XOR

SINDH TEXTBOOK BOARD

- A database is composed of tables, which store data in rows and columns.
- Each row is a record representing a single entity instance, and each column is a field representing an attribute.
- The primary key is used to uniquely identify each record, ensuring accuracy and preventing duplication.
- The Entity-Relationship (ER) Model provides a way to represent real-world systems using entities (objects), attributes (properties), and relationships (associations).
- ER diagrams visually illustrate how data elements are connected.
- Relationships can be:
  - One-to-One (1:1)
  - One-to-Many (1:M)
  - Many-to-Many (M:N) (resolved using a junction table)
- To maintain correctness across related tables, the concept of referential integrity is introduced.
- Referential integrity ensures that relationships remain valid by enforcing rules on foreign keys.
- Mechanisms such as cascade update and cascade delete automatically maintain consistency across tables.
- A relational schema acts as a blueprint of the database.
- Forms provide a user-friendly interface for entering, updating, and viewing data without directly interacting with tables.
- Queries are used to retrieve, filter, and analyze specific data from one or more tables based on defined criteria.
- Reports are used to present data in a formatted and organized manner, suitable for printing and sharing.

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Exercise

1. Encircle the correct answer:

i) Which of the following is a DBMS?
(a) MS Word (b) MS Excel
(c) MS Access (d) MS Paint

ii) In an ER Model, a rectangle represents:
(a) Attribute (b) Relationship
(c) Entity (d) Key

iii) A one-to-many relationship means
(a) One record relates to many (b) Many to one only
(c) One to One (d) None

iv) Which object is used to retrieve data?
(a) Table (b) Form
(c) Query (d) Report

v) Which function calculates average?
(a) Sum() (b) Count()
(c) Avg() (d) Max()

vi) Which key uniquely identifies each record in a table?
(a) Foreign Key (b) Primary Key
(c) Alternate Key (d) Candidate Key

vii) Which relationship requires a junction table?
(a) One-to-One (b) One-to-Many
(c) Many-to-Many (d) None

viii) Which database objects provides a user friendly interface for data entry?
(a) Query (b) Table
(c) Report (d) Form

ix) Which chart is best used to show trends over time?
(a) Pie Chart (b) Bar Chart
(c) Line Chart (d) Column Chart

x) Which query operation combines records into categories for summarization?
(a) Sorting (b) Filtering
(c) Group by (d) Validation

2. Fill in the Blanks:

i) In a relational database, a _____ key is used to establish a link between two tables by referring to the primary key of another table.

ii) If two entities are connected through an intermediate entity instead of directly, this usually indicates the resolution of a _____ relationship.

iii) When a record in one table is deleted, and related records in another table must also be removed automatically, this is controlled through _____ rules.

iv) A table created to resolve a many-to-many relationship is commonly referred to as a _____ table.

v) A candidate key that is not chosen as the primary key is known as _____.

3. Provide short descriptive answers of the following questions.

i) Differentiate between a field and a record.

ii) What is referential integrity?

iii) Explain the role of a Database Management System (DBMS).

iv) Explain a one-to-many relationship with an example.

v) Define data visualization and give one example.

SIDH TEXTBOOK BOARD

- The main components of IoT are sensors, actuators, connectivity, data processing, cloud computing, and user interface.
- IoT is implemented in smart cities, industry, healthcare, education, smart homes, and agriculture.
- In Pakistan, IoT can enhance traffic management, healthcare services, classroom administration, industrial production, and water-efficient agriculture.
- Data Analytics involves collecting, storing, cleaning, analyzing, and presenting data to extract valuable insights.
- Data Analytics supports schools, banks, hospitals, businesses, and government offices in making informed decisions.
- Information sources are divided into primary, secondary, and tertiary sources.
- Students should check information carefully before using it because not all online information is reliable.
- Computing has positive impacts, such as faster communication, easier learning, better healthcare, online banking, and improved farming.
- Computing also has negative impacts, including privacy risks, excessive screen time, job displacement, and overreliance on technology.
- Responsible use of computing means using technology safely, ethically, and wisely.
- Assistive technologies support individuals with disabilities in mobility, learning, communication, hearing, vision, and daily activities.
- These technologies promote equal access to education, healthcare, employment, and social participation.
- This unit helps students understand the connection between computing, daily life, society, career opportunities, and Pakistan's future development.

SINDH TEXT BOOK BOARD



1. Encircle the correct answer

i) A smartwatch records heart rate and sends it to a mobile phone. Which technology is mainly used here?

(a) Word Processing
(b) Internet of Things
(c) Printing Technology
(d) Analog Technology

ii) In a smart irrigation system, which device checks whether the soil is dry or wet?

(a) Actuator
(b) Sensor
(c) Monitor
(d) Smart hearing aid

iii) A school removes duplicate student records while preparing examination results. Which step of Data Analytics is this?
(a) Data collection (c) Data cleaning
(b) Data visualization (d) Data storage

iv) Which of the following is a primary source of information?
(a) Textbook (c) Dictionary
(b) Interview (d) Summary website

v) A student uses an unknown website without an author name or references. This source is likely to be:
(a) Reliable (c) Unreliable
(b) Primary (d) Official

vi) Which technology supports people with hearing, vision, learning, or movement difficulties?
(a) Gaming technology (c) Assistive technology
(b) Manufacturing technology (d) Antivirus technology

vii) Which of the following is an actuator in a smart water system?
(a) Soil moisture sensor (c) Temperature sensor
(b) Water valve (d) Camera sensor

viii) Which type of source usually provides summarized information for quick reference?
(a) Primary source (c) Tertiary source
(b) Secondary source (d) Unverified source

ix) Which method is used to present data through charts and graphs?
(a) Data collection (c) Data cleaning
(b) Data storage (d) Data visualization

x) Which assistive technology converts on-screen text into speech?
(a) Screen reader (c) Motion sensor
(b) Smart meter (d) Water pump

2. Fill in the Blanks

i) A _____ detects changes such as temperature, movement, light, or soil moisture.

ii) An _____ performs a physical action, such as opening a valve or switching on a fan.

iii) Correcting duplicate, missing, or incorrect records is called data _____.

iv) An interview is an example of a _____ source of information.

v) Screen readers and hearing aids are examples of _____ technologies.

SIDH TEXTBOOK BOARD

Summary

- Digital literacy involves searching carefully, checking source reliability, collecting evidence, and communicating findings responsibly.
- A strong project begins with a focused research question and a suitable plan for using primary and/or secondary data.
- Interviews, surveys, prototypes, simulations, and observation each produce different kinds of evidence and should be chosen thoughtfully.
- Sampling, bias, privacy, consent, and citation all affect the quality and ethics of student research.
- Data should be cleaned and organized before analysis so that tables, charts, and conclusions are accurate.
- Different visuals serve different purposes: tables show exact values, bar charts compare categories, line graphs show trends, and infographics summarize key findings quickly.
- A strong digital artefact connects the research question, method, findings, conclusion, and references in a clear structure.
- Results should be communicated differently for different audiences while remaining accurate, readable, and evidence-based.

SIDH TEXT BOOK BOARD

Exercise

1. Encircle the correct answer

i) Which of the following best describes digital literacy?
(a) The ability to type quickly using a keyboard
(b) The ability to use digital tools to find, evaluate, create, and communicate information responsibly
(c) The ability to install software on a computer
(d) The ability to use social media for entertainment

ii) A student asks classmates, "How does mobile phone use affect your concentration during study?" What type of data is mainly collected?
(a) Quantitative data because it can be counted easily
(b) Secondary data because classmates already know the answer
(c) Qualitative data because it collects opinions and experiences
(d) Numerical data because mobile phone use is related to time

iii) Which search query is most suitable for finding a reliable PDF report about internet usage in Pakistan?
(a) internet usage Pakistan
(b) best internet use in Pakistan social media
(c) "internet usage in Pakistan" filetype:pdf
(d) Pakistan internet all websites latest

iv) Which option is the best example of primary data?
(a) A government report downloaded from the internet
(b) A research article written by another author
(c) Survey responses collected directly from Class XI students
(d) Statistics copied from an online database

v) Why are spreadsheets useful when presenting research data?
(a) They automatically make the research question correct
(b) They help organize data, perform calculations, and create charts
(c) They replace the need for collecting original data
(d) They ensure all online information is reliable

vi) A digital artefact created after research should mainly:
(a) repeat long paragraphs from websites
(b) communicate results and conclusions clearly
(c) include decoration only
(d) avoid any data visuals

vii) Data collected directly by the researcher is called:
(a) Secondary data (b) Processed data
(c) Primary data (d) Digital data

viii) Which search operator helps search within a specific website or domain?
(a) OR (b) AND
(c) NOT (d) site:

ix) Copying someone else's work without giving credit is called:
(a) Copyright (b) Referencing
(c) Citation (d) Plagiarism

x) Asking permission before collecting responses is part of research:
(a) Analysis (b) Sampling
(c) Presentation (d) Ethics

2. Fill in the Blanks

i) A well-defined _____helps a student choose suitable data-collection methods and stay focused during research.

ii) Data collected directly by the researcher through surveys, interviews or observation is called _____data.

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